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UMM AL-QURA UNIVERSITY

DATA COLLECTION AND PROCESSING

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DATASET LINK

https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset

AMAZON SALES ANALYSIS PROJECT LINK

https://colab.research.google.com/drive/1pviippB1OHf5tmn_3TQsBW9frfIStcJv?usp=sharing

INTRODUCTION

The Amazon Sales dataset represents a rich and structured collection of product-related attributes, including pricing information, discount levels, and customer-generated ratings. This project aims to move beyond basic data preprocessing by providing a comprehensive analytical framework that extracts meaningful insights .from the dataset

Initially, the data was carefully cleaned and preprocessed by handling missing values, removing irrelevant features, and ensuring data consistency. Exploratory Data Analysis (EDA) was then conducted to investigate the distribution of product ratings .and to evaluate whether these ratings accurately reflect customer satisfaction

Furthermore, the analysis explored the relationship between key pricing variables, such as actual price and discounted price, and their impact on product performance. Special attention was given to rating count and rating credibility in order to assess .the reliability and trustworthiness of customer feedback

To deepen the analysis, correlation techniques and dimensionality reduction methods, including Principal Component Analysis (PCA), were applied to identify the .most influential features and uncover hidden patterns within the dataset

The findings of this study contribute to a better understanding of customer behavior, product quality perception, and pricing strategies, ultimately enabling more informed decision-making in e-commerce environments.

INTRODUCTION & DATA OVERVIEW

Dataset Description

Source: The dataset is a structured collection of Amazon sales information obtained from Kaggle .

Size: The dataset consists of 1,465 rows and 16 columns.

Type: Currently, all 16 columns are identified as object data types (strings) in the initial loading. This is due to the presence of non-numeric characters like currency symbols (₹) and percentage signs (%) within numeric fields.

Data Representation (Variables of the Data)

The dataset contains the following 16 variables, all of which are initially loaded as object type :

1. product_id: Unique identifier for each product .
2. product_name: Full name of the product.
3. category: Product classification path.
4. discounted_price: Price after discount (includes currency symbols)
6.actual_price: Original retail price
7.review-_content: The full text of the customer's review.
8.rating: Product score (stored as text)
9.rating_count: Number of users who rated the product
10.about_product: Product description and features
11.user_id: Unique identifier for the reviewer
12.user_name: Name of the reviewer
13.review_id: Unique ID for the review
14.review_title: Short title of the review
15.img_link: URL of the product image
16.product_link: URL of the product page on Amazon1

Objective of the Analysis

The project aims to clean and preprocess the raw data to extract meaningful insights regarding:

Customer Satisfaction: Evaluating how product ratings are distributed

Pricing Strategy: Analyzing the relationship between price, discounts, and product performance

Rating Reliability: Using rating_count and credibility features to identify trustworthy products

Feature Correlation: Identifying influential factors using techniques like Correlation and PCA

Dataset Selection and Relevance

Real-World Context: This dataset provides a realistic look at e-commerce trends and consumer behavior

Data Cleaning Opportunity: Many columns are currently stored as object types (strings) and require conversion to numerical formats for analysis

Multidimensional Insight: Combining text (reviews) and numbers (prices/ratings) allows for a deep, comprehensive study of market quality

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating
count	1465	1465	1465	1465	1465	1465	1465
unique	1351	1337	211	550	449	92	28
top	B083342NKJ	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹999	50%	4.1
freq	3	5	233	53	120	56	244

ATA QUALITY ASSESSMENT

Data Quality Issues Identification

- **Missing Values**: After running df.isnull().sum(), it was observed that only the rating_count column contains 2 missing entries, while all other columns are complete

Duplicate Records: The dataset was checked for identical rows using df.duplicated().sum(), and the result was 0, indicating no duplicate data

Inconsistent Labels: By examining unique values in the category column, it was found that labels are inconsistently structured, containing long strings of hierarchical categories separated by symbols (e.g., "|")

```
Computers&Accessories|Accessories&Peripherals|Cables&Accessories|Cables|USBCables',
Computers&Accessories|NetworkingDevices|NetworkAdapters|WirelessUSBAdapters',
Electronics|HomeTheater,TV&Video|Accessories|Cables|HDMICables',
Electronics|HomeTheater,TV&Video|Televisions|SmartTelevisions',
```

Invalid Values: A logic check was performed for numeric ranges (Rating between 0-5, Discount between 0-100, and no negative values). The results confirmed 0 invalid or impossible values across all checked columns

Detection Methods

df.isnull().sum(): Used to identify and count missing values in each column

df.duplicated().sum(): Applied to verify the uniqueness of each record and ensure no redundancy

df['category'].unique(): Used to display all unique labels in the category column to detect inconsistencies in naming conventions

Impact on Analysis

Data Completeness: Missing values in rating_count prevent a full representation of product popularity and can lead to gaps in statistical summaries

Calculation Barriers: Storing numerical data as object types is a critical issue; it prevents any mathematical operations, such as calculating average prices or identifying correlations

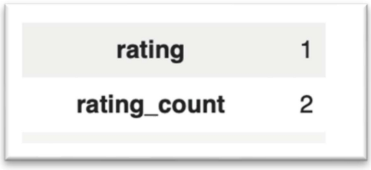
Clarity and Grouping: The inconsistent and overly long category labels make it difficult to group products effectively or create clear, readable visualizations

DATA CLEANING PROCESS

Converting Data Types

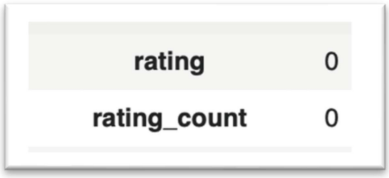
We removed unwanted symbols such as commas
and percentage signs, then converted the columns (discounted_price, actual_price, discount_percentage, rating, and rating_count) into numeric format using pd.to_numeric(). This step was necessary because the columns were originally stored as text (object type), which prevents proper numerical analysis and calculations. As a result, all relevant columns became numeric and ready for further analysis.

Handling Missing Values



rating	1
rating_count	2

We handled missing values in the rating and rating_count columns using the median. Two methods were considered for handling missing values: deletion and median imputation. Deletion involves removing rows with missing values; however, this approach may lead to loss of important data, especially when the dataset is not very large. Therefore, we chose median imputation because it is less affected by outliers and provides a better representation of the central tendency of the data. This approach allowed us to preserve the dataset while effectively handling missing values.



rating	0
rating_count	0

After: All missing values were filled, and the dataset became complete without losing data.

—

Removing Duplicates

We applied the drop_duplicates() function to check for and remove any duplicate rows in the dataset. This step was necessary because duplicate records can bias analysis results and lead to incorrect insights. After applying this method, no duplicate rows were found in the dataset; however, performing this step ensured the quality and reliability of the data.

—

Cleaning Category Column

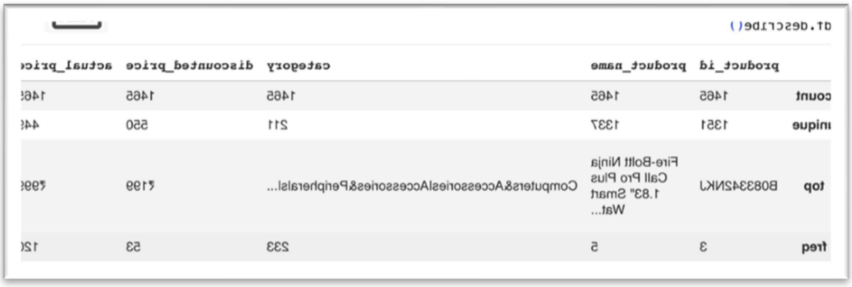
We simplified the category column by splitting the values using the "|" symbol and keeping only the main category. This step was necessary because the original values were complex and hierarchical, which made the analysis more difficult. After cleaning, the categories became simpler and more consistent, making the dataset easier to analyze and interpret.

Before cleaning

product_id	object
product_name	object
category	object
discounted_price	object
actual_price	object
discount_percentage	object
rating	object
rating_count	object
about_product	object
user_id	object
user_name	object
review_id	object
review_title	object
review_content	object
img_link	object
product_link	object

```
# Data we displayd before handling outliers
# تم عرض البيانات قبل التعامل مع القيم الشاذة
df.describe()
```

	discounted_price	actual_price	discount_percentage	rating	rating_count
count	1465.000000	1465.000000	1465.000000	1465.000000	1465.000000
mean	3125.310874	5444.990635	47.691468	4.096587	18277.634812
std	6944.304394	10874.826864	21.635905	0.291574	42727.398216
min	39.000000	39.000000	0.000000	2.000000	2.000000
25%	325.000000	800.000000	32.000000	4.000000	1191.000000
50%	799.000000	1650.000000	50.000000	4.100000	5179.000000
75%	1999.000000	4295.000000	63.000000	4.300000	17325.000000
max	77990.000000	139900.000000	94.000000	5.000000	426973.000000



After cleaning:

product_id	object
product_name	object
category	object
discounted_price	float64
actual_price	float64
discount_percentage	int64
rating	float64
rating_count	float64
about_product	object
user_id	object
user_name	object
review_id	object
review_title	object
review_content	object
img_link	object
product_link	object
mid_category	object

```
df.describe()
```

	discounted_price	actual_price	discount_percentage	rating	rating_count
count	1465.000000	1465.000000	1465.000000	1465.000000	1465.000000
mean	1472.065823	3130.301556	47.691468	4.107850	11343.800683
std	1508.698409	3196.451132	21.635905	0.253277	13309.244942
min	39.000000	39.000000	0.000000	3.550000	2.000000
25%	325.000000	800.000000	32.000000	4.000000	1191.000000
50%	799.000000	1650.000000	50.000000	4.100000	5179.000000
75%	1999.000000	4295.000000	63.000000	4.300000	17325.000000
max	4510.000000	9537.500000	94.000000	4.750000	41526.000000

```
df.head()
```

product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating
07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories	399.0	1099.0	64	4.2
098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories	199.0	349.0	43	4.0
96MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories	199.0	1899.0	90	3.9
08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories	329.0	699.0	53	4.2
08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories	154.0	399.0	61	4.2

DATA TRANSFORMATION & FEATURE ENGINEERING

Importance of Data Transformation

Data transformation was necessary to ensure that all features are in a consistent format and scale, which improves model performance and prevents bias. The dataset originally contained features with different ranges (rating vs. rating count), categorical variables, and data not properly structured for machine learning, which could negatively affect model performance. In addition, categorical variables needed to be transformed into numerical format, as machine learning models cannot process textual data directly.

- Before Data Transformation

	discounted_price	discount_percentage	rating	rating_count	rating_credibility
count	1465.000000	1465.000000	1465.000000	1465.000000	1465.000000
mean	1472.065823	47.691468	4.107850	11343.800683	33.896466
std	1508.698409	21.635905	0.253277	13309.244942	8.421366
min	39.000000	0.000000	3.550000	2.000000	3.900074
25%	325.000000	32.000000	4.000000	1191.000000	28.529638
50%	799.000000	50.000000	4.100000	5179.000000	35.197006
75%	1999.000000	63.000000	4.300000	17325.000000	40.448231
max	4510.000000	94.000000	4.750000	41526.000000	50.511971

The statistical summary shows a clear imbalance in feature scales. Variables such as discounted_price and rating_count have very large ranges and high standard deviations, while rating is limited to a very narrow range. For example, rating_count reaches over 41,000, whereas rating varies only between 3.55 and 4.75. This significant variation can lead to biased model learning, where features with larger magnitudes dominate the model. Therefore, data transformation—particularly scaling—is necessary to standardize the features and ensure fair contribution from all variables.

Scaling and Normalization

The statistical summary reveals a significant variation in feature ranges. Variables such as *discounted_price* and *rating_count* span large values with high standard deviation, while *rating* is confined to a very small range. This imbalance can negatively impact model performance, as features with larger magnitudes may dominate the learning process, especially in models sensitive to scale such as Linear Regression and PCA.

Given that the dataset consists of continuous numerical features, **StandardScaler** was chosen as the most appropriate method. It standardizes each feature by centering it around a mean of 0 and scaling it to a standard deviation of 1. This ensures that all features contribute equally to the model, improves numerical stability, and allows algorithms that rely on distance or variance to perform effectively. Normalization methods such as MinMaxScaler were not used, as StandardScaler is more suitable for data with varying distributions and for models sensitive to variance, such as PCA.

- After Scaling

	discounted_price	discount_percentage	rating	rating_count	mid_category	rating_credibility
0	-0.711496	0.754029	0.363955	0.971476	Computers&Accessories	1.010979
1	-0.844105	-0.216911	-0.425963	2.268536	Computers&Accessories	1.026304
2	-0.844105	1.956146	-0.820922	-0.256736	Computers&Accessories	0.132902
3	-0.757909	0.245441	0.363955	2.268536	Computers&Accessories	1.278941
4	-0.873943	0.615323	0.363955	0.417988	Computers&Accessories	0.830590

After applying Standard Scaling, all numerical features (such as *discounted_price*, *discount_percentage*, *rating*, and *rating_count*) are transformed to have values centered around zero with a standard deviation close to one. This is evident from the presence of both positive and negative values, indicating that the data has been normalized relative to the mean. As a result, all features now share a similar scale, preventing any single feature from dominating the model.

Encoding Techniques for Categorical Variables

Categorical variables, specifically *mid_category*, were encoded using One-Hot Encoding with the function:
pd.get_dummies(X, columns=['mid_category'], drop_first=True)
This method converts each category into a separate binary column represented as True/False (equivalent to 1/0), allowing machine learning models to process categorical data numerically. One-Hot Encoding was chosen because *mid_category* is a nominal variable with no inherent order, and using other methods such as Label Encoding could introduce misleading ordinal relationships. Alternative techniques such as Ordinal Encoding are suitable only for ordered categories, while Target Encoding may lead to data leakage if not handled carefully. Therefore, One-Hot Encoding was the most appropriate and safe choice for this dataset. Additionally, the parameter drop_first=True was applied to avoid multicollinearity (dummy variable trap), ensuring that the encoded features remain independent and suitable for modeling.

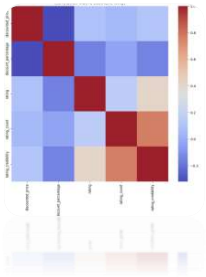
Feature Engineering Techniques Applied

Creation of a new feature (rating_credibility)

A new feature was engineered by combining *rating* and *rating_count* to better represent the reliability of product ratings.

Feature Selection using Correlation

Correlation analysis was used to evaluate relationships between features
Based on the results, *actual_price* was removed due to its strong correlation with *discount_percentage*, as this could lead to redundancy and multicollinearity in the model.



Based on the correlation matrix, *rating_credibility* shows a strong positive relationship with *rating*, while *rating_count* has a moderate positive relationship. In contrast, *discounted_price* has a weak positive relationship, and *discount_percentage* shows a weak negative relationship, indicating limited influence on the target.

The feature *actual_price* was removed because it is highly correlated with *discount_percentage*, which leads to redundancy and multicollinearity. Removing it helps improve model performance and reduces unnecessary complexity.

3. **Feature Selection using RFE (Recursive Feature Elimination)**

Feature Selection using RFE (Recursive Feature Elimination) is a technique used to identify and retain the most important features for the model. It works by training a model (in this case, Linear Regression), evaluating feature importance, and recursively removing the least significant features until the desired number of features is reached.

Initially, the dataset contained 11 features (7 categorical and 4 numerical). RFE was applied to select the top 7 most relevant features based on their contribution to predicting the target variable (*rating*).

The main goal of this technique is to reduce dimensionality, eliminate redundant or less informative features, and improve model performance, interpretability, and efficiency.

Number of features before RFE: 11

Number of features after RFE: 7

Selected features:

- 'mid_category_Computers&Accessories',
- 'mid_category_Home&Kitchen', 'mid_category_OfficeProducts',
- 'mid_category_Electronics', 'mid_category_HomeImprovement',
- 'mid_category_Toys&Games', 'rating_credibility'**]**

Dimensionality Reduction using PCA

PCA was applied to reduce dimensionality while preserving most of the variance. Initially, the model considered 7 components, where the explained variance ratios were:

Explained Variance:

[0.6159, 0.1994, 0.1661, 0.0166, 0.0012, 0.0005, 0.0002]

Cumulative Variance:

[0.6159, 0.8153, 0.9815, 0.9981, 0.9993, 0.9998]

The first principal component alone explains approximately **61.6%** of the total variance, indicating that a large portion of the information is captured in a single dimension. Using two components increases the explained variance to approximately **81.53%**, which represents most of the data. Additionally, using three components captures more than **98%** of the total variance.

Based on this, the dimensionality was reduced to 2 principal components, as they provide a good balance between reducing complexity and preserving information. This significantly simplifies the model while maintaining high data representation and performance.

The Importance of the New Feature for the Model and for Analysis

The newly created feature (*rating_credibility*) is useful because it combines both the rating value and the number of reviews, providing a more reliable measure of product quality. Instead of relying on the rating alone, this feature accounts for how many users contributed to the rating, which helps reduce bias from products with high ratings but very few reviews. As a result, it improves the model's ability to make more accurate and meaningful predictions.

Impact of Data Transformations on Data Quality and Analytical Performance

Data transformations significantly improved both data quality and analytical performance. First, scaling the numerical features using StandardScaler ensured that all variables were on a similar scale, preventing features with large values (such as *rating_count*) from dominating the model. This not only improved model fairness and stability but also enhanced data consistency by bringing all features into a comparable range, reducing the impact of extreme values and improving numerical reliability.
Second, encoding categorical variables using One-Hot Encoding converted non-numeric data into a machine-readable format, allowing the model to effectively utilize category information without introducing false relationships. This transformation improved data quality by eliminating ambiguity in categorical representation and ensuring that each category is clearly and independently defined, which enhances interpretability and reduces the risk of misleading patterns.

Final Transformed Results Summary

After applying all data transformations, the dataset became fully prepared for modeling. All numerical features were standardized using StandardScaler, resulting in values centered around zero with a unit variance. Categorical variables were successfully encoded into binary columns, making them suitable for machine learning algorithms.
Highly correlated and redundant features, such as *actual_price*, were removed to improve data quality and reduce multicollinearity. Feature selection techniques further reduced the dataset to the most relevant features, decreasing the number of features from 11 to 7.
Finally, PCA was applied to reduce dimensionality, where the data was represented using only 2 principal components while preserving approximately 81.53% of the total variance.
Overall, the final dataset is clean, consistent, reduced in complexity, and optimized for efficient and accurate model performance.

- After Scaling

	discounted_price	discount_percentage	rating	rating_count	mid_category	rating_credibility
0	-0.711496	0.754029	0.363955	0.971476	Computers&Accessories	1.010979
1	-0.844105	-0.216911	-0.425963	2.268536	Computers&Accessories	1.026304
2	-0.844105	1.956146	-0.820922	-0.256736	Computers&Accessories	0.132902
3	-0.757909	0.245441	0.363955	2.268536	Computers&Accessories	1.278941
4	-0.873943	0.615323	0.363955	0.417988	Computers&Accessories	0.830590

- After Encoding

	mid_category_Computers&Accessories	mid_category_Electronics	mid_category_Health&PersonalCare	mid_category_Home&Kitchen
	True	False	False	False
	True	False	False	False
	True	False	False	False
	True	False	False	False
	True	False	False	False

The final transformed dataset reflects the successful application of scaling and encoding techniques. After scaling, all numerical features (such as *discounted_price*, *discount_percentage*, *rating*, and *rating_count*) are standardized, with values centered around zero and distributed across positive and negative ranges. This confirms that the data is now on a consistent scale and suitable for modeling.
After encoding, the categorical variable *mid_category* was transformed into multiple binary columns represented as True/False values, where each column indicates the presence or absence of a specific category. This transformation enables the model to interpret categorical information effectively.
Overall, the transformed dataset is now fully numerical, well-structured, and ready for analysis and machine learning tasks.

ANALYSIS RESULTS

The analysis phase was conducted after completing all preprocessing steps, including data cleaning, outlier handling, feature engineering, and data transformation. This ensured that the dataset used for analysis was reliable, consistent, and suitable for extracting meaningful insights.

Initially, the dataset exhibited variability in numerical features, particularly *discounted_price*, *actual_price*, and *rating_count*. As shown in the descriptive statistics before preprocessing, some variables contained extreme values where the maximum values were significantly higher than the median, indicating the presence of outliers.

Outlier detection was performed using box plot visualization.

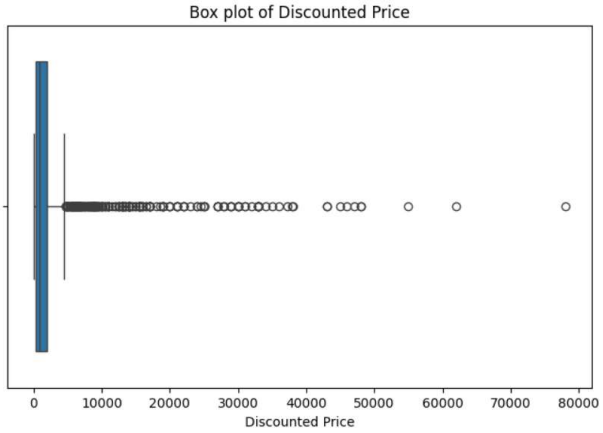


Figure 1: Box Plot of Discounted Price

As shown in Figure 1, several outliers are present in the *discounted_price* variable. These values were not removed or modified, as they represent realistic variations in product pricing rather than data errors. Retaining these values ensures that the dataset reflects real-world conditions.

From the data quality assessment, it was observed that missing values were minimal and limited only to the *rating_count* column, with two missing entries. These values were handled using median imputation, which is suitable for skewed data and ensures that the dataset remains complete without introducing bias.

Moreover, no duplicate records or invalid values were detected, confirming that the dataset is structurally consistent and free from redundancy.

A key enhancement in the analysis was the creation of the *rating_credibility* feature. This feature combines both *rating* and *rating_count*, capturing not only how high a product is rated but also how reliable that rating is. This is important because ratings with a higher number of reviews are generally more trustworthy than those with fewer reviews.

The correlation analysis provided further insight into relationships between variables.

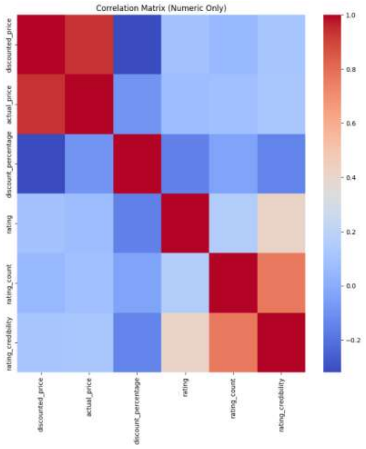


Figure 2: Correlation Matrix after Feature Engineering

The correlation analysis was conducted after introducing the new feature (rating_credibility). The results show that rating_credibility has a strong positive correlation with rating, indicating that it is a highly informative feature for predicting customer satisfaction.

Additionally, rating_count shows a moderate positive relationship with rating, while discounted_price and actual_price exhibit a strong correlation with each other, confirming redundancy between these variables.

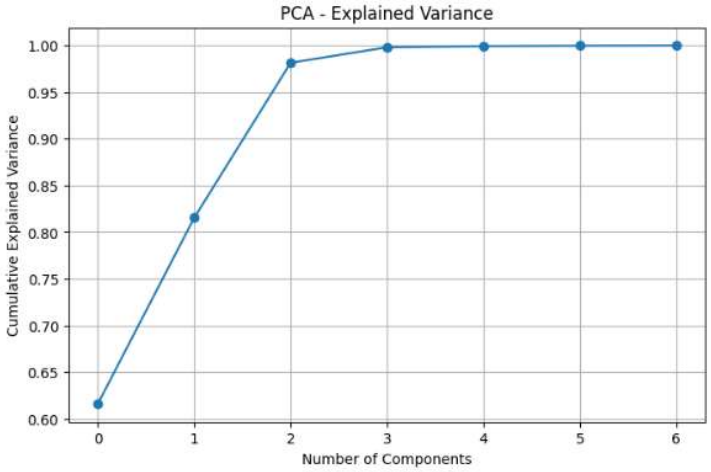
In contrast, price-related features have a weak relationship with rating, suggesting that pricing alone does not strongly influence customer satisfaction.

To improve model efficiency, Recursive Feature Elimination (RFE) was applied, reducing the number of features from 11 to 7.

```
Number of features before RFE: 11
Number of features after RFE: 7
Selected features: ['mid_category_Computers&Accessories',
'mid_category_Home&Kitchen', 'mid_category_OfficeProducts',
'mid_category_Electronics', 'mid_category_HomeImprovement',
'mid_category_Toys&Games', 'rating_credibility']
```

The selected features include key numerical variables such as *discounted_price*, *discount_percentage*, *rating_count*, and *rating_credibility*, along with relevant categorical features derived from *mid_category*.

Principal Component Analysis (PCA) was applied to understand the structure of the dataset.



The results indicate that the first principal component explains approximately 61% of the variance, while the first two components explain around 81.5%. Using three components increases the explained variance to approximately 98%, indicating that most of the dataset's information can be represented using a small number of dimensions.

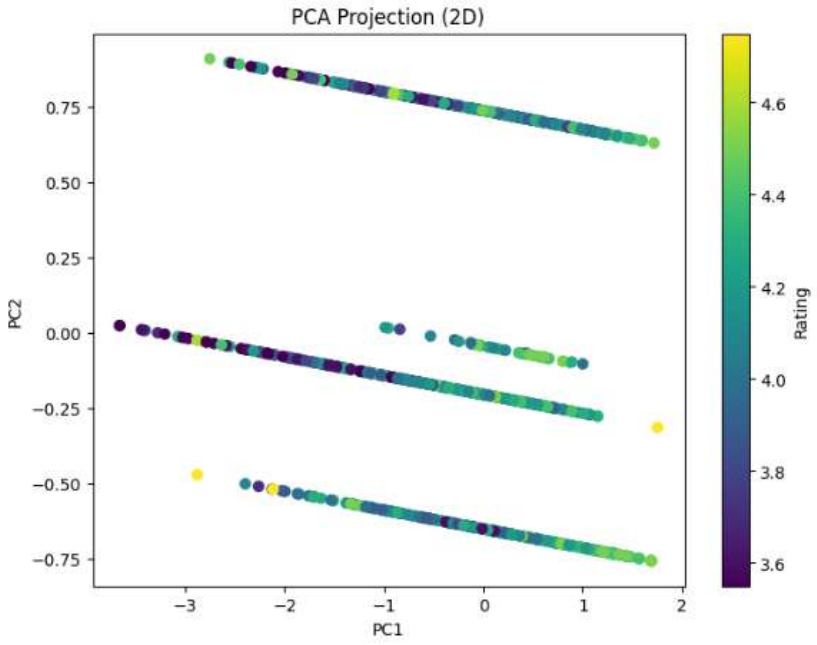


Figure 5: PCA Visualization

The PCA projection shows structured patterns in the data, suggesting relationships between variables. However, the distribution of ratings does not form distinct clusters, indicating that ratings are influenced by multiple factors rather than a single dominant variable.
Overall, the analysis shows that customer ratings are generally high, but are more strongly influenced by review-related features than by price alone.

CONCLUSIONS

The analysis results indicate that product ratings are more strongly influenced by category-related features and review-based factors than by price. In particular, mid_category, rating_credibility, and rating_count show the strongest relationship with the target variable (rating), while price-related features such as discounted_price and discount_percentage have a weaker effect.

Furthermore, feature selection using RFE confirmed that the most relevant features include both key numerical variables and selected categorical features, improving model efficiency while preserving important information.

Additionally, PCA results show that most of the data variance can be captured using a small number of components, indicating that the dataset can be effectively represented in a lower-dimensional space.

Overall, these findings suggest that customer engagement and product category play a more significant role in shaping ratings than pricing factors alone.

Limitations

Despite the data cleaning and preprocessing steps
applied, some limitations remain in the dataset. First, certain values were originally stored in inconsistent formats, which required conversion and may have introduced minor inaccuracies during the cleaning process. In addition, handling missing values using median imputation may not fully reflect the true distribution of the data in all cases.

Furthermore, the dataset may still contain hidden inconsistencies or noise that were not detected during preprocessing. Finally, the analysis was limited to the available features in the dataset, and the absence of additional relevant variables may affect the depth of insights. Future work could include collecting more comprehensive data, applying advanced data cleaning techniques, and exploring additional features to improve analysis accuracy.

REFERENCES

https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset